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Rb-Sr study of impact glass and country rocks from the Henbury meteorite crater field

W. COMPSTON and S. R. TAYLOR
Department of Geophysics and Geochemistry, Australian National University, Canberra, A.C.T. 2600, Australia

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Abstract—Rb–Sr data for the Henbury country rock (Winnall Beds) and the laterally equivalent Pertatataka Formation 95 miles ENE of Henbury, fall close to an isochron with age 730 m.y. ± 45 m.y. and an Sr^{87}/Sr^{86} initial ratio of 0·724. The scatter about the isochron is interpreted as variation in initial Sr^{87}/Sr^{86} reflecting small changes in Sr provenance. The impact glass data are particularly well-aligned over a 15% range in Rb/Sr, and independently define an isochron of 675 ± 100 m.y. Certain of the sediments will fit the glass alignment to within experimental error. These data show that no selective loss of Rb relative to Sr occurred during melting. Comparison of Rb, Sr abundances of glasses and sediments suggests that, although similar, none of the analyzed subgreywackes are identical with the inferred parent material of the glasses, which would contain a greater clay fraction. Alternatively the composition of the glasses was biassed by selective melting of the clay matrix.

INTRODUCTION

THE formation of glass from country rock during meteorite impact provides an opportunity to study the chemistry of such processes, with both the parental material and the end product being available for comparison. Such work is of relevance to the chemistry of tektites, formed during analogous, but larger impact events. A particularly good example exists at Henbury, Northern Territory, Australia, (24°35′S, 133°09′E) where the impact of an iron meteorite shower has produced at least 12 craters. Impact glass formed from subgreywacke type sandstones in which the main crater has been excavated. Studies of the major and trace element composition of the impact glass have been made by TAYLOR and KOLBE (1964, 1965) and TAYLOR (1966, 1967). These studies have shown that

1. The glass is similar in composition across the strewn field.
2. Fe, Ni, Co and the Pt group elements have been added from the impacting meteorite in variable amounts.
3. The closest similarity to the glass in chemistry is with subgreywacke sandstones outcropping in the walls of the northwest member (7A) of the double main crater (No. 7) and with a sample from an outthrown flap of sediment. No glass of quartzite composition has been identified.
4. The glass shows overall similarities in trace element composition to the subgreywackes. In comparison with the sediments analysed, it contains more Rb, Cs, Cu, Cr, Mg which "probably indicates a slightly higher clay content in the parent material of the glasses." (TAYLOR and KOLBE, 1965, p. 751.)
5. Among the volatile elements, Na is lower in the glass. Li and Pb were equivalent and Rb, Cs and Cu showed enrichment in the glass compared with the

sediment. TAYLOR and KOLBE (1964, 1965) interpreted this as limiting selective volatilization to a minor role, within the limits of analytical error, and attributed the compositional variations as due to minor differences between the analysed sediments and the actual source rock.

These studies have involved major and trace element comparisons, with limits of ± 5–10% precision for the trace element analytical data.

Precise Rb–Sr isotopic determinations offered the possibility of studying the behaviour of these elements in more detail than previously. Rb is of particular interest because of possible volatilization, since KOLBE *et al.* (1967) interpreted the Sr^{87}/Sr^{86}–Rb^{87}/Sr^{86} plots for the Bosumtwi crater sediments and the Ivory Coast tektites as indicating selective loss of Rb during melting. Accordingly, a selected suite of glass and sediments from Henbury was analysed.

PREIMPACT GEOLOGY AT HENBURY

The craters are excavated in Winnall Beds, a sandy lateral equivalent of the shaly Pertatataka Formation of Upper Proterozoic age. The strata strike East–West, and dip South at 35°. There is a gradual change in lithology and chemical composition of the sedimentary rocks from north to south across the crater field. The most northerly exposed strata are subgreywackes with 77% SiO_2 occurring as blocks at the north wall of the main crater. To the south, thin bands of impure quartzites appear, interbedded with the subgreywackes. Higher in the sequence both the number of quartzite bands and the overall SiO_2 content of the sediments increase until, south of the main crater, a low ridge occurs composed of impure quartzites. These have an SiO_2 content of 92–94%. The detailed petrography of these samples has been given by TAYLOR and KOLBE (1965).

CRATER FIELD

The iron meteorite was a medium octahedrite (Fe: 91·95; Ni: 7·45; Co: 0·60%. Analyst H. B. Wiik).

The structural geology of several of the craters (6, 7, 8, 10) has been described by MILTON (1968). Descriptions of the craters have been given by ALDERMAN (1933) and photographs of the craters have been published by HODGE (1965). The age of the craters has been given at less than 4700 years, based on the cosmogenic C^{14} content of the meteorite (KOHMAN and GOEL, 1963).

Craters 1–8 are excavated in dominantly soft subgreywacke sediments. The intersection of harder bands produces the looped rays described by MILTON and MICHEL (1965). Craters 10–13 occur in the harder quartzites of the ridge.

ANALYSED SAMPLES

The locations of the analysed samples are shown by TAYLOR (1967; Fig. 2) and are listed in Appendix 2 of the same paper where they are correlated with MILTON'S (1968) units. Eight glass samples from different areas of the strewnfield and thirteen sedimentary rock samples, both from strata exposed in the crater walls, and across the strewnfield were analysed. Sample HB 27 is from an outthrown flap of sediment 100 meters northwest of the rim crest. In the preimpact geological

setting, this sample was probably close to the centre of Crater 7A. The variations in sedimentary rock types across the crater field are useful in localising the site of glass formation, since the glass has a very restricted composition in comparison with the sediments. The ultimate limits of such comparisons are determined by variations in chemistry and lithology between the strata involved in glass formation, and their lateral equivalents in the crater walls, a distance of about 150 feet. Disturbance of the strata by impact, as described by MILTON (1968) places additional limits on the precise identification of the parent material. The correction for meteoritic iron, cobalt and nickel adds a further restriction since there are difficulties in estimating the meteoritic contribution (commonly of several percent). The major and trace element chemistry point to sediments within crater 7A and to MILTON's unit D as the probable source of the glass.

Unpublished analyses of three argillites from the Pertatataka Formation in the Ooraminna No. 1 Bore, some 100 miles to the east of the Henbury crater site are also relevant. They were made previously in this laboratory by V. M. BOFINGER to set limits to the age of the Pertatataka Formation.

ANALYTICAL METHODS

The data for Rb, Sr and Sr^{87}/Sr^{86} are given in Table 1. Rb and Sr for the sediments 16, 17, 18, 20, 21, 8, 9, 19 and soil 27A were determined by X-ray fluorescence spectrography (NORRISH and CHAPPELL, 1967), and the remainder by standard isotope-dilution methods. All results are precisely related to the values shown in Table 1 for the NBS standard K-feldspar 70A. Sr^{87}/Sr^{86} was measured either directly without tracer or during isotope dilution with a single Sr^{84} tracer, and is normalised to 8·375 for Sr^{88}/Sr^{86}. The coefficient of variation for Rb^{87}/Sr^{86} by both the X-ray and isotope-dilution methods is approximately 0·5%, and the standard error in Sr^{87}/Sr^{86} about $\pm 0{\cdot}0003$.

The Ooraminna Bore data (obtained in 1964) have comparable internal precision but may be systematically biased relative to the others by up to $\pm 1\%$ in Rb^{87}/Sr^{86}, owing to tracer variations. Their Sr^{87}/Sr^{86} were calculated from double Sr^{86}, Sr^{84} isotope-dilution measurements; they are less precise (standard error 0·001) and have been corrected for a bias of 0·15%.

ISOCHRON INTERPRETATION OF GLASSES AND SEDIMENTS

The eight glasses are particularly well-correlated on the isochron-diagram, Fig. 1. The small range in Rb^{87}/Sr^{86} (about 15%) is generated by independent variation of 13 to 16% in concentration of both Rb and Sr, so that the observed Rb distribution is an essential part of the correlation with Sr^{87}/Sr^{86}. Interpreting the latter as an isochron, the age of the glasses is quite well-defined at 675 ± 100 m.y., which obviously cannot be distinguished from the isotopic character of the Henbury and Ooraminna sediments (Fig. 1). Rb volatilization during the formation of the glasses would have displaced their points to the left of the sedimentary isochron in Fig. 1, parallel to the Rb^{87}/Sr^{86} axis. Clearly, this has not occurred.

The good correlation between the Henbury and Ooraminna Bore sediments in Fig. 1 is consistent with their stratigraphic equivalence. Despite the different names of the various specimens (argillite, subgreywacke, quartzite) the clay–chlorite matrix which would carry most of the Rb and Sr, is probably very similar for all specimens. The matrix shows virtually no diagenetic alteration, so that the sedimentary isochron is probably equal to or older than the age of deposition. Rb in the matrix would be mainly authigenic but Sr probably mainly inherited.

Table 1. Rb, Sr and Sr^{87}/Sr^{86} for sediments of the Winnall Beds at the Henbury Crater site, of the Pertatataka Formation from the Ooraminna No. 1 Bore, and for the Henbury Impact glasses

		Rb (ppm)	Sr (ppm)	Rb^{87}/Sr^{86}	Sr^{87}/Sr^{86}
Sediments, Henbury Crater Site					
3,	subgreywacke	116·9	63·4	5·36	0·7788
4,		74·8	52·6	4·13	0·7679
5,		119·0	68·2	5·07	0·7749
16,		113·9	61·9	5·34	0·7757
17,		117·7	66·9	5·11	0·7743
18,		115·1	65·8	5·08	0·7730
20,		108·8	67·6	4·64	0·7708
21,		133·1	75·3	5·13	0·7727
27,		151·0	75·1	5·85	0·7826
27A,	soil	83·9	78·9	3·08	0·7448
6,	quartzite	52·9	38·8	3·95	0·7651
8,		36·8	69·3	1·537	0·7435
9,		61.1	45·0	3·94	0·7613
19,		52·6	33·4	4·57	0·7774
Sediments, Ooraminna No. 1 Bore					
GA 1059A, argillite		218·2	64·3	9·75	0·8244
GA 1059F, argillite		195·0	69·5	8·06	0·8078
GA 1059E, argillite		161·6	77·8	5·97	0·7853

Henbury Crater Glasses

Specimen	Rb* (ppm)	Sr* (ppm)	Correction factor*	Rb^{87}/Sr^{86}	Sr^{87}/Sr^{86}
1	146·9	87·3	1·057	4·89	0·7700
2	161·1	97·0	1·114	4·81	0·7699
11	152·7	101·2	1·014	4·38	0·7653
28	142·5	91·8	1·015	4·51	0·7667
29	153·7	97·6	1·036	4·54	0·7672
31	146·1	99·6	1·027	4·26	0·7644
1213	148·2	95·0	1·078	4·50	0·7669
1415A	148·1	96·4	1·019	4·46	0·7660
NBS Standard K-feldspar 70a					
	530·0	65·1			1·200

*Corrected for dilution by meteoritic iron as described in text.

Sr^{87}/Sr^{86} in the detritus will remain constant only as long as the processes and nature of the provenance are constant, so that some variation in the initial Sr^{87}/Sr^{86} of the Henbury sediments must be expected.

The distribution of points in Fig. 1 is quite consistent with the above interpretation. The three argillites 1059 are well-correlated and could be fitted to the Henbury samples 3, 27, 5 and 6 without much scatter in excess of experimental error. Presumably the initial Sr^{87}/Sr^{86} for each of these was approximately the same and the age of deposition about 725 m.y. (taking λRb^{87} as $1{\cdot}39 \times 10^{-11}$ yr^{-1}). On this interpretation the initial Sr^{87}/Sr^{86} in the quartzites 19 and 8 would be higher and that in the glasses and remaining Henbury samples lower. However it is difficult to be objective in making such groupings, and further geochemical and mineralogical work is proceeding on the same specimens in the hope of finding some

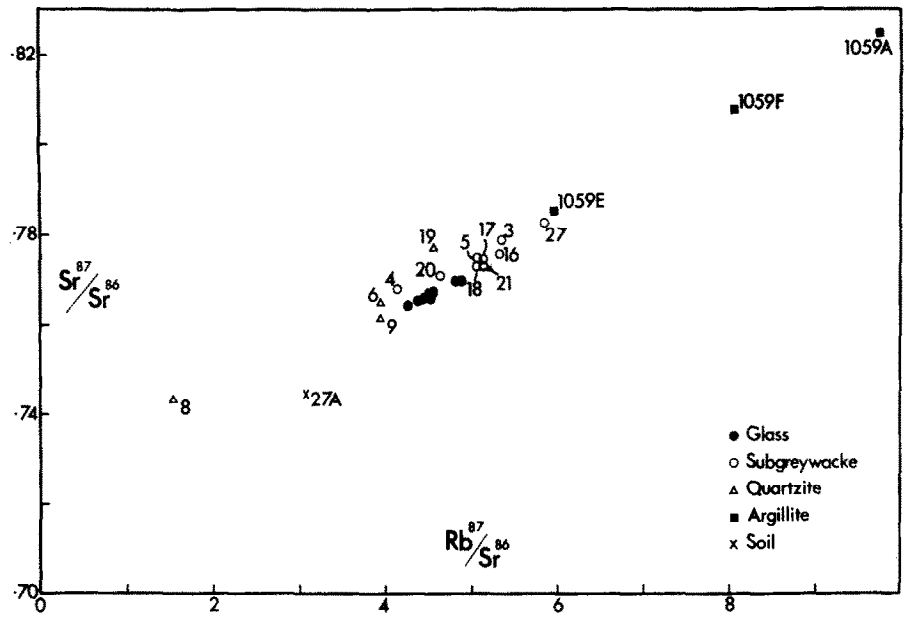


Fig. 1. Isochron-diagram showing data for sediments and glasses from the Henbury crater area, and for stratigraphically-equivalent sediments 1059A, E and F from the Ooraminna No. 1 Bore, 95 miles E.N.E. of Henbury.

property by which these small differences in Sr provenance may be identified. In the meantime, no precise estimate of the age of deposition can be given.

It is possible however to test statistically which of the Henbury sediments will fit the alignment shown by the glasses with the same precision. The group comprising the glasses, quartzite 9 and subgreywackes 18, 16, 21 and 17 will fit a 735 $\pm$ 45 m.y. isochron to within experimental error (initial Sr^{87}/Sr^{86} 0·720 $\pm$ 0·003). The inclusion of subgreywacke 27 slightly exceeds experimental error, and it is predictable that the other near point, specimen 5, will also do so. This test, of similarity or otherwise in Sr provenance, can be added to comparisons of major- and trace-element compositions in the search for the particular sedimentary stratum at Henbury from which the glasses were formed.

Comparative Rb, Sr Abundances in Glasses and Sediments

The Rb and Sr concentrations in Table 1 each have a coefficient of variation of less than 0·5%, as compared with 5–10% by emission spectrography. A much closer comparison can now be made between the glasses and sediments than attempted previously by Taylor and Kolbe (1965), and the correction for meteoritic contamination in the glasses becomes a more important factor.

During glass-formation, the Henbury iron-meteorite (92 wt.% metallic Fe), diluted SiO_2 and increased total iron in the parental sediment by variable amounts; in SiO_2–FeO co-ordinates, such displacements from the parent compositions have a calculable gradient. We have combined this fact with the inverse SiO_2-total Fe correlation observed in the (unaltered) Henbury sediments (Taylor and Kolbe, 1965) to make correction for meteorite contamination. SiO_2 and FeO for the parent

sediment are taken as the point of intersection of the displacement-line drawn through the measured SiO_2, FeO point of the glass, with the above SiO_2–FeO correlation line. It is thus inferred that on a meteorite-free basis, the eight analyzed glasses have SiO_2 contents ranging from 75·9 to 70·3% and FeO from 4·4 to 5·5%, in contrast to the correction used by TAYLOR (1967) of a constant 4·0% FeO, and a higher average SiO_2 content. The chemistry of the glasses including Rb and Sr has therefore been adjusted by the factors, inferred SiO_2/observed SiO_2, which are given in Table 1.

Figure 2 shows the strong inverse correlation between SiO_2 and Rb for the Henbury sediments and the glasses. However only one sediment, 27, falls into

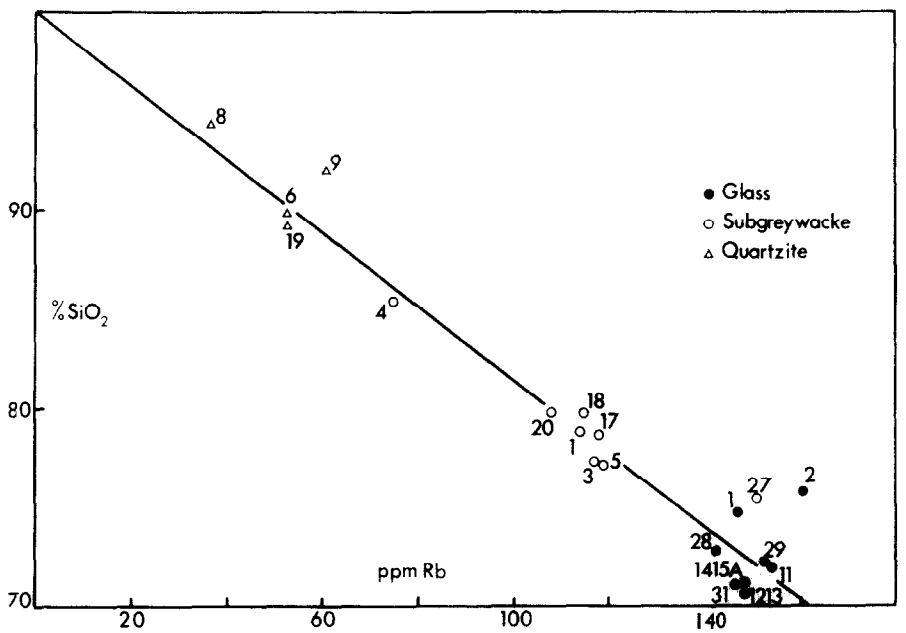


Fig. 2. The inverse correlation of Rb with SiO_2 for the Henbury sediments and glasses, showing possible bias in the latter towards higher Rb.

the SiO_2–Rb field of the glasses: the others, including those which most resemble the glasses in Rb^{87}/Sr^{86} and Sr provenance, all have lower Rb. A similar bias in the subgreywackes occurs for Sr, K, Fe, Al—all the elements in fact which are concentrated in the clay–chlorite matrix. If the main group of six subgreywackes in Fig. 2 is taken as the average parental material, then the size of the bias is about 25%.

A possible bias was remarked previously by TAYLOR and KOLBE (1965), who suggested that it might be due to a slightly higher clay content in the specific parental rocks. This may be so. However bias may also occur through the selective melting of the matrix relative to its quartzose component. Obviously the question of bias depends completely on a reliable correction of the glass-chemistry for meteorite dilution. This problem is currently being re-examined.

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